AtomStorage · C0

Give bytes stable identity. Own their acceptance.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A library accession desk: it assigns an immutable reference and records which items were actually accepted.

Why it matters: Consumers can retry, swap providers and rebuild indexes only if reference identity and accepted-source order remain explicit.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

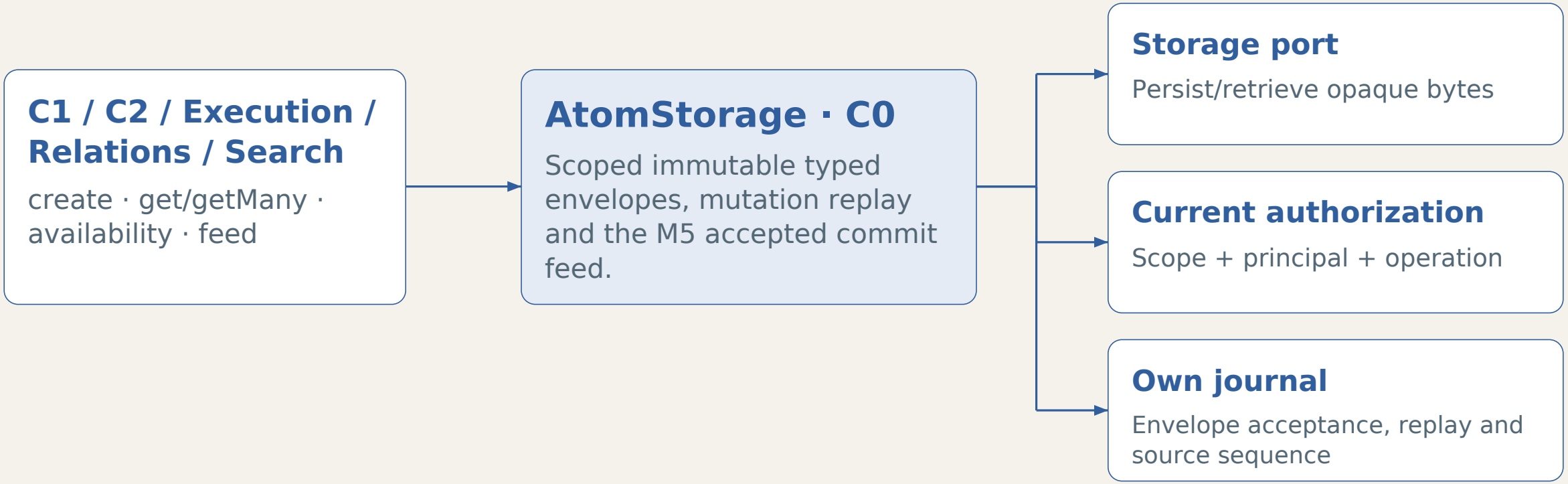
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Scoped immutable typed envelopes, mutation replay and the M5 accepted commit feed.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Envelope + replay identity

Immutable reference, content type and request semantics

Accepted source feed

Source UUID, scope and contiguous sequence in the M5 profile

OUTSIDE THIS RESPONSIBILITY

Knowledge meaning

No query interpretation, belief admission or relation acceptance

Cross-owner transaction

DA bytes may commit before envelope acceptance; clients recover through ports

Historical immutable-bytes/1; fresh immutable-commit-feed/1 · contract 1.1.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
create	Accept one scoped immutable envelope	Same intent replays; changed intent conflicts
get / getMany	Retrieve exact typed bytes for references	Live authority and integrity checks
availability	Report selected provider evidence	Preserve provider/durability distinctions
head / readCommits	Enumerate accepted source prefix	Fresh immutable-commit-feed/1 profile

M5 caps 512 accepted atoms and 4096 bytes/atom; legacy limits remain profile-specific.

A worked example through the boundary

M5 REAL DeTTa + OWNER JOURNAL · one deliberate feed-acceptance kill

01 · STEP

Persist bytes

Write through the opaque storage port.

02 · STEP

Accept

Commit envelope and feed metadata together.

03 · STEP

Lose reply

Kill after acceptance, before caller acknowledgment.

04 · STEP

Replay

Recover one reference and one source sequence.

What the example establishes

Accepted identity is authoritative; indexing is derived.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Immutable identity

An accepted reference binds scope, revision and exact content.

I2

Replay semantics

Same principal/scope/key with changed meaning conflicts.

I3

Contiguous accepted feed

Acceptance and sequence metadata commit together; orphans do not enter the feed.

I4

Provider neutrality

The consumer does not inspect concrete provider internals.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Bytes persist; acceptance not yet committed	Retry recovers bytes, then completes owner acceptance.	Independent commit boundary
Acceptance commits; reply lost	Replay returns the same logical acceptance.	M5 feed kill case
Provider substitution	Select an independent same-profile implementation on a fresh store.	No populated-state migration

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

2 × 2

Providers × bindings

M1.1/M1.2 workflow substitution matrix

1

M5 feed SIGKILL case

Acceptance before response

512 / 4096 B

M5 atoms / per-atom limit

Contract bounds, not demonstrated scale capacity

The substitution matrix establishes matching logical outcomes. Existing whole-workflow timings include storage, messaging, workers and retries; they are not AtomStorage latency.

PROPOSED BENCHMARK · NOT RUN

Proposed: create/get/getMany across provider, payload and batch size; separate cold open, warm read and replay, with p50/p95 plus storage amplification.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	DeTTa-backed AtomOwner and fresh FeedAtomOwner with their own journals.
FIXTURE / LOCAL TEST POLICY	Independent atomic-file AtomStorage and feed; generated bytes and hostile ports.
BLOCKED / UNSUPPORTED	No required M5 gate blocked. Replacement of an already populated namespace is unsupported.
UNRUN	Power loss, distributed consistency, pruning/compaction and live namespace migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Quantify provider overhead	Fixed corpus and equivalent durability labels	Separate lookup latency from DA reconstruction
Design retention/pruning	Source-prefix and cursor invalidation contract first	No silent holes in accepted history
Add a production replacement	Independent owner transitions + conformance	Same consumer, explicit profile, no hidden migration

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Accepted identity is authoritative; indexing is derived.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m5.py --gates storage --environment <env>
--output <fresh>
```

Build pinned native providers first. Storage gate uses real DeTTa; fixture passes cannot substitute for it.

`fusion/atoms.py`
`fusion/atom_feed.py`
`fusion/adapters/fixture_feed.py`
`docs/module-interfaces/m5-contracts.md`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.